

## 0.1 $p$ -Groups

**Theorem 0.1.1.** Let  $G$  be a group of order  $p^n$  for some prime  $p$  and  $n \in \mathbb{N}$ . Then,  $G$  has a normal series  $\{P_i\}_{i=0}^n$  s.t

$$1 = P_0 \triangleleft P_1 \triangleleft \cdots \triangleleft P_{n-1} \triangleleft P_n = G$$

where  $|P_i| = p^i$  for each  $i$ .

*Proof.* First, we recall that for any  $p$ -group  $G$ , the center  $Z(G) \neq 1$ . This follows from the class equation:

$$p^n = |G| = |Z(G)| + \sum |G : C_G(g_i)| = |Z(G)| + \sum p^{r_i} \quad (1 \leq r_i \leq n-1).$$

This implies  $p$  divides  $|Z(G)|$ , so  $|Z(G)| > 1$ .

Now, we use induction on  $n$ . For  $n = 2$ ,  $|G/Z(G)| = p$  or  $1$ . In either case,  $G/Z(G)$  is cyclic, which implies  $G$  is Abelian. By Cauchy's Theorem,  $G$  has a subgroup of order  $p$ , and the claim is satisfied.

For  $n \geq 2$ , suppose the claim is true for groups of order  $p^n$ . Let  $|G| = p^{n+1}$ . Since  $1 \subsetneq Z(G)$ , by Cauchy's Theorem, there exists a normal subgroup  $N \triangleleft G$  such that  $N \leq Z(G)$  and  $|N| = p$ . Now,  $G/N$  is a group of order  $p^n$ . By the induction hypothesis,  $G/N$  has a normal series:

$$1 = N_0/N \triangleleft N_1/N \triangleleft \cdots \triangleleft N_{n-1}/N \triangleleft N_n/N = G/N.$$

The th Isomorphism Theorem gives  $N_{n-1} \triangleleft G$  and  $|N_{n-1}| = p^n$ . Thus, by induction, the claim is proved.  $\square$

**Lemma 0.1.2.**  $G/Z(G)$  is cyclic  $\iff G$  is Abelian.